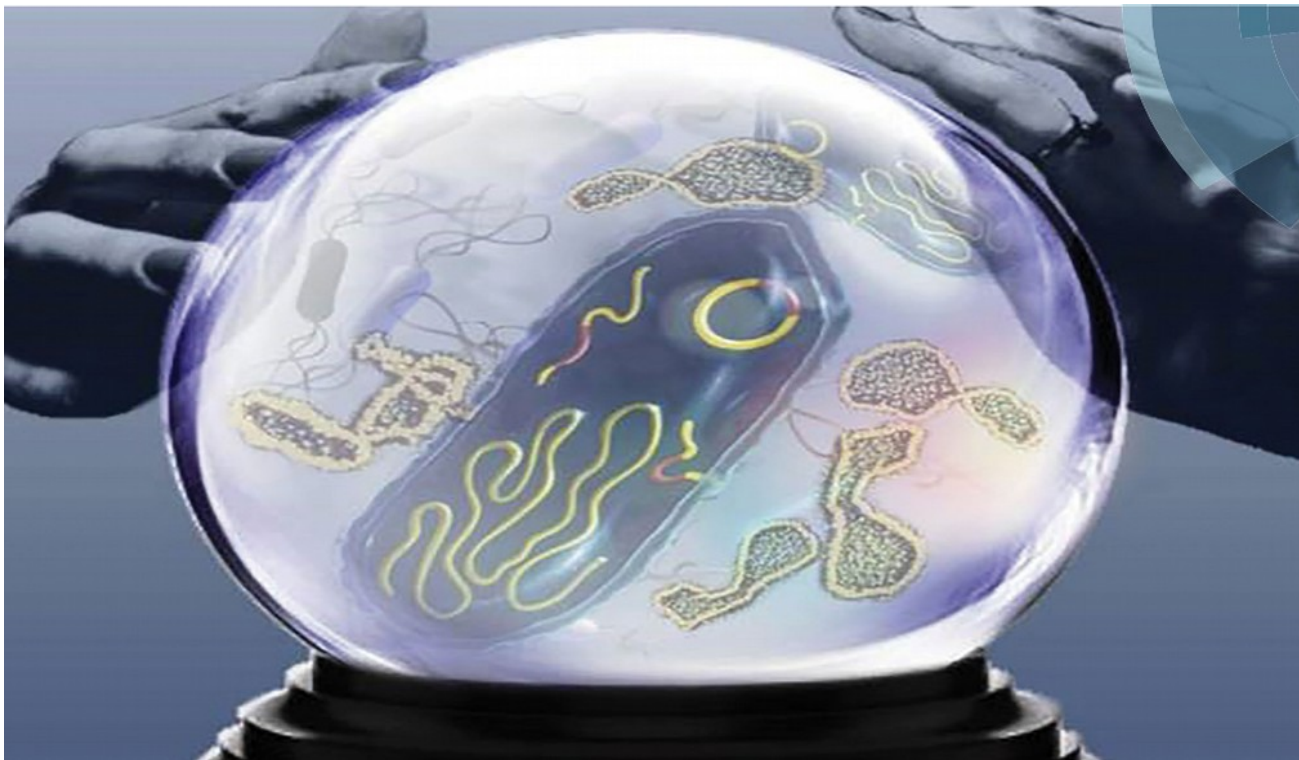


Introduction: Machine Learning MultiOmics Integration

Physalia course

Nikolay Oskolkov, MRG Group Leader, NIRI, Riga, Latvia



@NikolayOskolkov



@osolkov.bsky.social

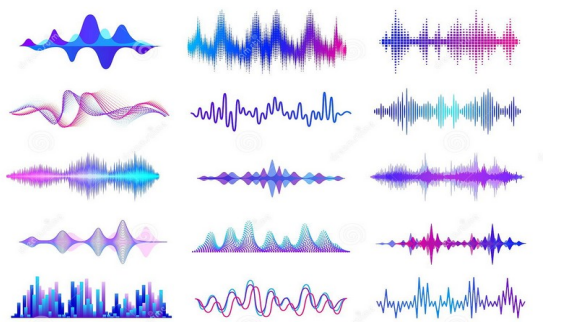


Personal homepage:
<https://nikolay-osolkov.com>

Introduction: High Dimensional Omics Data

Tabular

Sound



Image



Text

Editing Wikipedia articles on Medicine

Engage with editors

Part of the Wikipedia experience is meeting and responding to feedback from other editors. Do not submit your content on the last day. Check back on your article. Read recent comments from the Wikipedia community will likely read and respond to it, and it would be better for you to acknowledge the time they volunteer to public your work. Everything submitted to Wikipedia is reviewed by multiple, not human, but they are not a consensus, but if you do, please acknowledge it.

Be accurate

You're creating a resource millions of people use to make medical decisions, so it's really important to be accurate. Wikipedia is used more for medical information than the websites for WebMD, NIH, and the WHO. But with great power comes great responsibility!

Understand the guidelines

Wikipedia editors in the medical area have developed additional guidelines to ensure that the content on Wikipedia is medically sound. Take extra time to read and understand these guidelines. When you edit an article, ensure your changes meet these special requirements. If not, your work is likely to be undone by other editors as they clean up after you. Then take valuable volunteer time away from creating content. If you're not comfortable working under these guidelines, talk to your instructor about an alternative off-wiki assignment.

Watch out for close paraphrasing

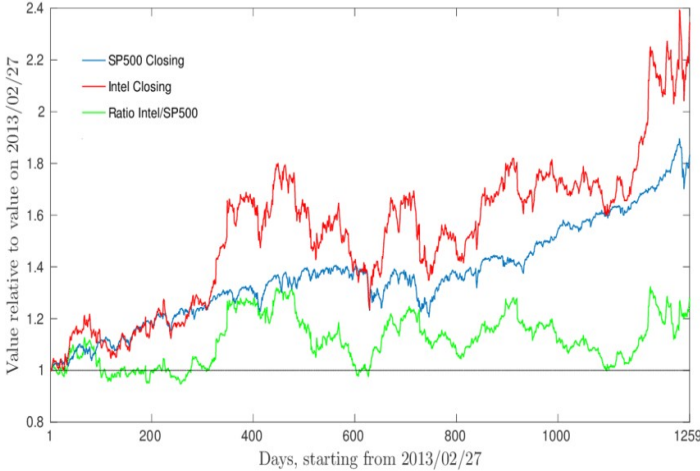
Paraphrasing or close paraphrasing is never okay on Wikipedia and is a violation of your university's academic honor code. It's even worse on Wikipedia, as valuable volunteer time that could be used to create good content is instead used to clean up plagiarized work.

Scared? Don't be!

Everybody on Wikipedia wants to make the best encyclopedia they can. Take the time to understand the rules, and soon you'll be contributing to a valuable resource you use on a daily basis!

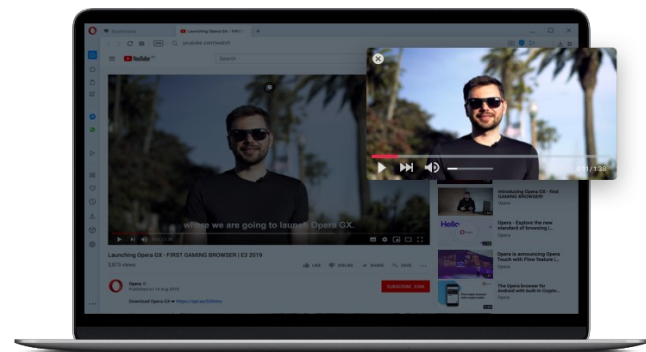
Time Series

Normalized Financial Time Series and Ratio

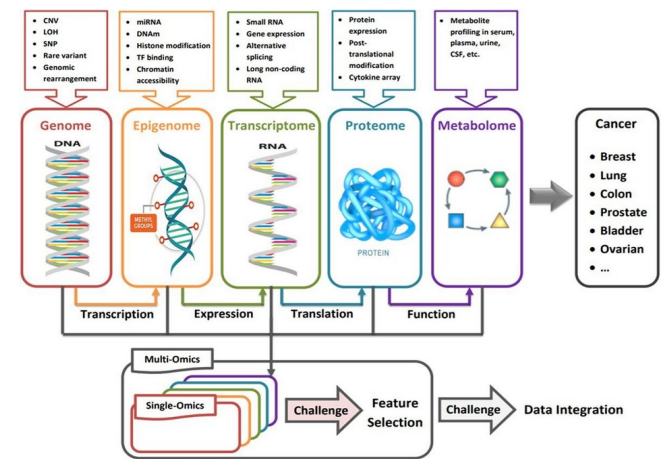


DATA

Video



Omics Data



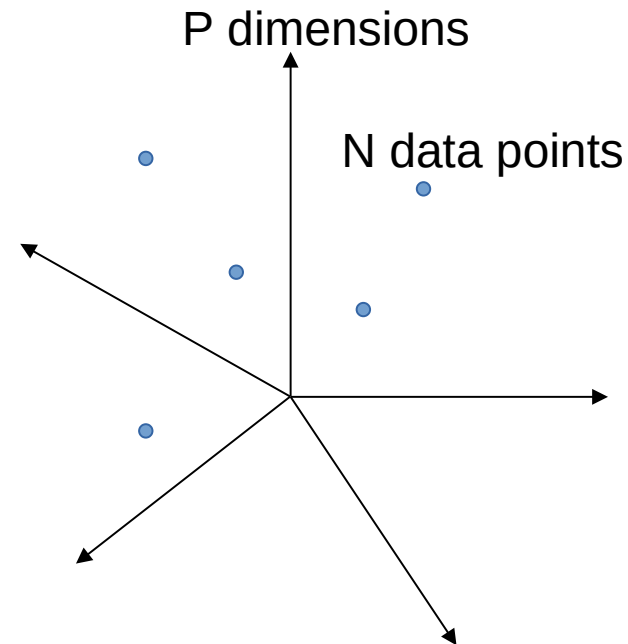
Statistical observations:
e.g. samples, cells etc.

Features: genes, proteins,
microbes, metabolites etc.

N

0	3	1	0	2	3	8	1	1	3
1	1	0	0	7	1	2	2	3	3
1	2	2	0	0	6	7	1	2	2
1	2	3	10	0	4	6	1	0	5
3	2	2	1	4	3	2	1	6	0
7	4	4	5	3	9	6	1	6	1
7	1	1	5	2	8	9	1	3	6
5	0	1	6	2	0	0	0	1	5
1	6	3	3	4	6	2	0	1	1
1	2	2	4	1	1	3	0	8	2

P



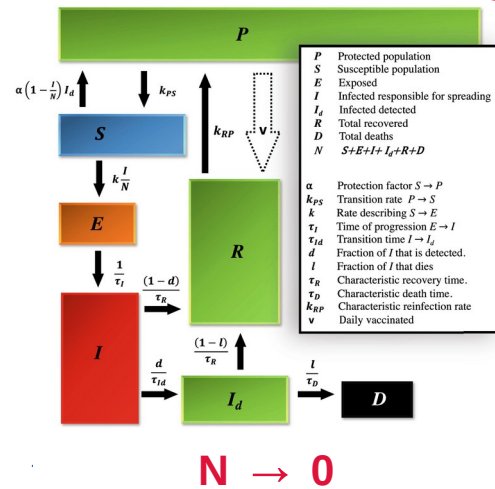
High Dimensional Data:
 $P \gg N$

For a robust statistical analysis, one should properly “sample” the P-dimensional space, hence large sample size is required, $N \gg P$

P is the number of features (genes, proteins, genetic variants etc.)
 N is the number of observations (samples, cells, nucleotides etc.)

Biology / Biomedicine

Mathematical modeling



Bayesianism



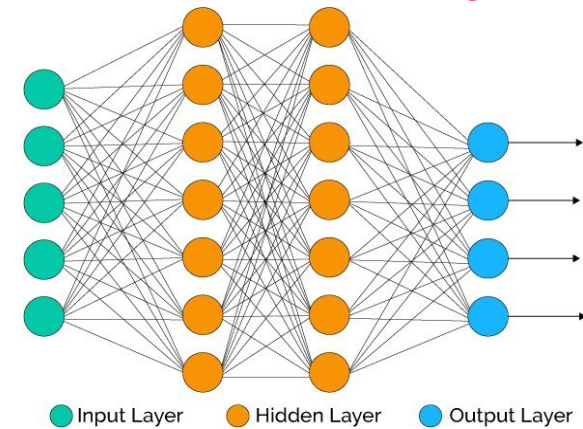
$N \ll P$

Frequentism



$N \approx P$

Machine Learning



$N \gg P$

Hypothesis-driven

Amount of Data

Data-driven

The Curse of Dimensionality

Ex.1

$$Y = \alpha + \beta X$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$(X^T X)^{-1} \sim \frac{1}{\det(X^T X)} \cdots \rightarrow \infty, \quad n \ll p$$

Ex.2 $E[\hat{\sigma}^2] = \frac{n-p}{n} \sigma^2$

Biased ML variance estimator in HD-space

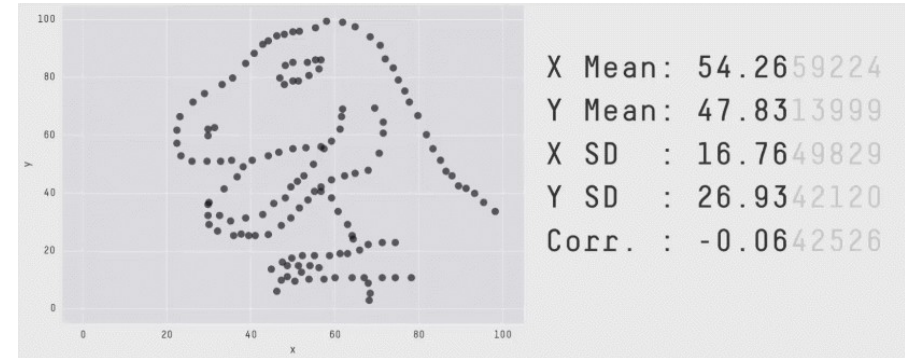
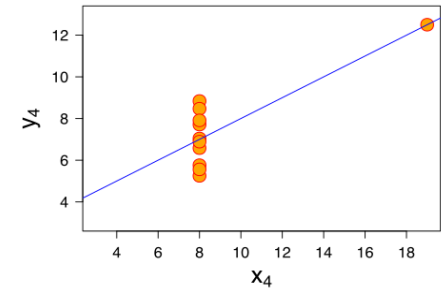
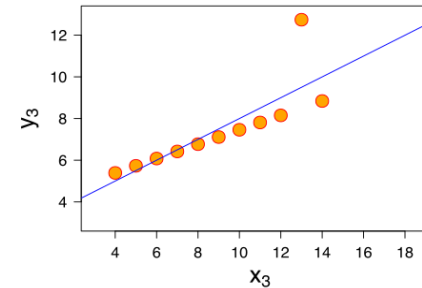
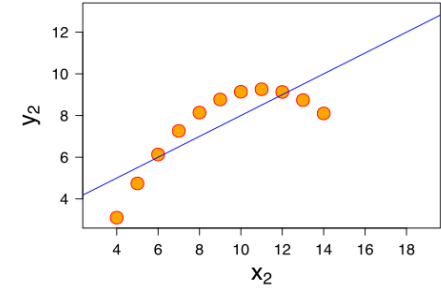
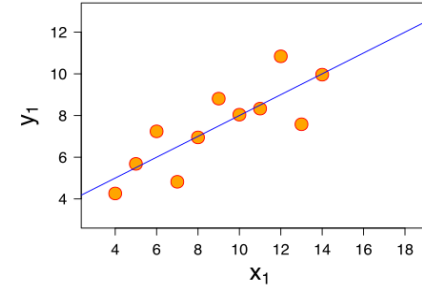
- Maximum likelihood based
- Focus on summary statistics
- Focus too much on p-values

$$L(x_i | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp - \frac{\sum_{i=1}^N (x_i - \mu)^2}{2\sigma^2}$$

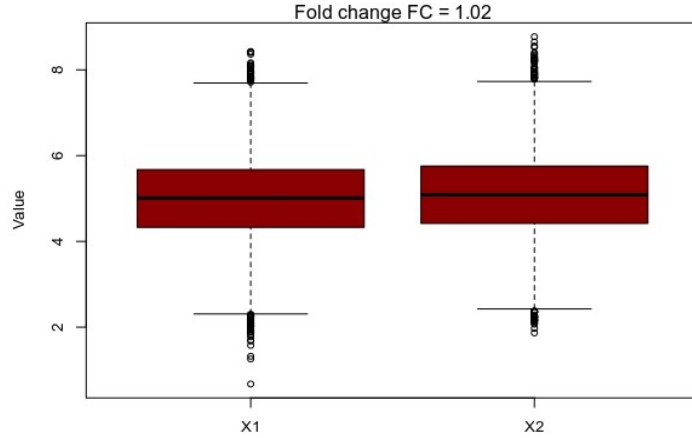
$$\frac{\partial L(x_i | \mu, \sigma^2)}{\partial \mu} = 0; \quad \frac{\partial L(x_i | \mu, \sigma^2)}{\partial \sigma^2} = 0$$

$$\mu = \frac{1}{N} \sum_{i=0}^N x_i - \text{mean estimator}$$

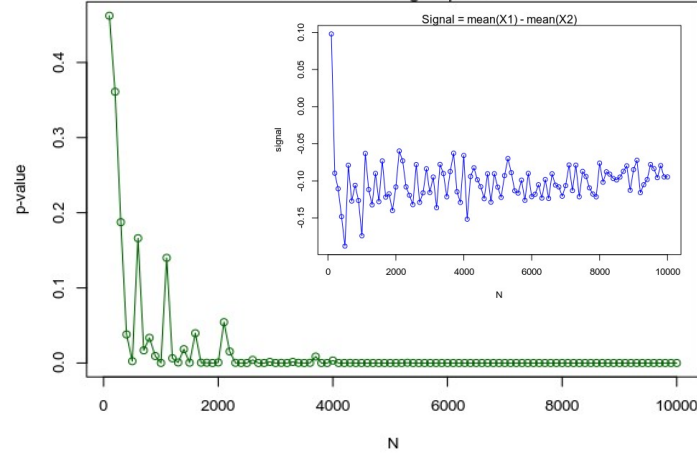
$$\sigma^2 = \frac{1}{N} \sum_{i=0}^N (x_i - \mu)^2 - \text{variance estimator}$$



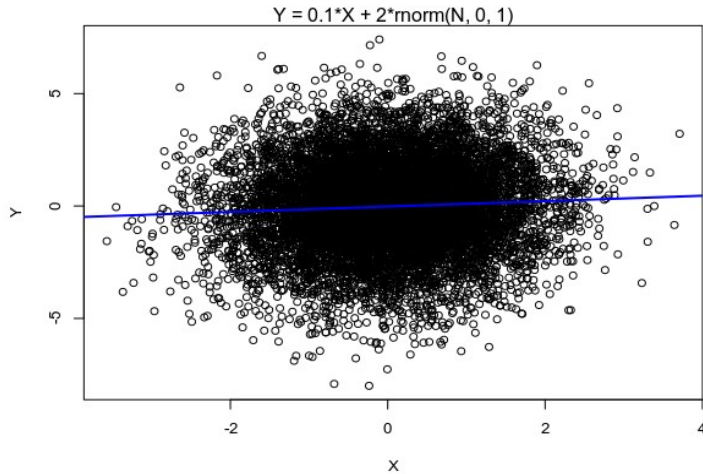
t-test statistics as functions of sample size



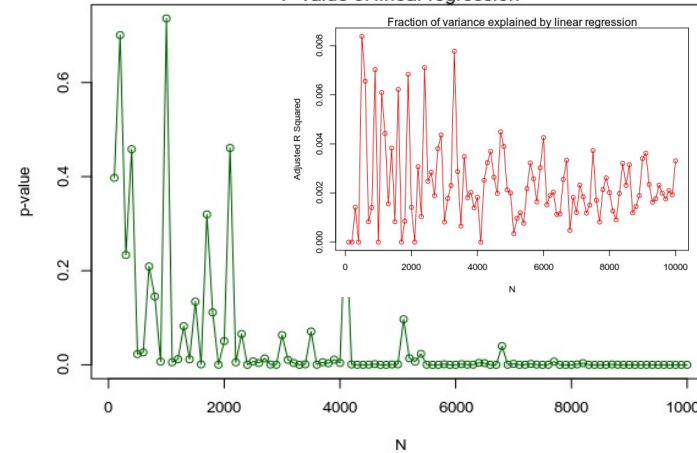
P-value of two-group t-test



Linear regression statistics as functions of sample size



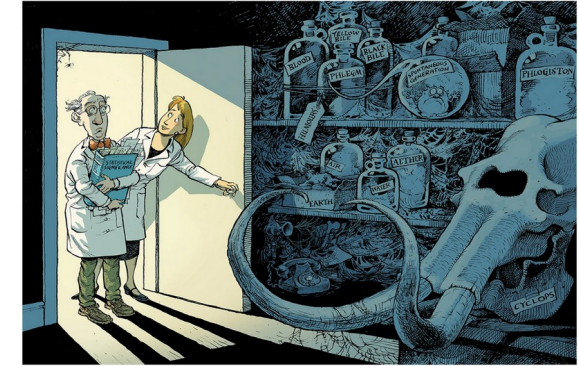
P-value of linear regression



Scientists rise up against statistical significance

Valentin Amrhein, Sander Greenland, Blake McShane and more than 800 signatories call for an end to hyped claims and the dismissal of possibly crucial effects.

Valentin Amrhein Sander Greenland & Blake McShane



It is questionable whether p-value is the best metric for ranking features

```
1 n <- 20 # number of samples
2 p <- 2 # number of features / dimensions
3 Y <- rnorm(n)
4 X <- matrix(rnorm(n * p), n, p)
5 summary(lm(Y ~ X))
```

Call:
lm(formula = Y ~ X)

Residuals:

	Min	1Q	Median	3Q	Max
	-2.0522	-0.6380	0.1451	0.3911	1.8829

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.14950	0.22949	0.651	0.523
X1	-0.09405	0.28245	-0.333	0.743
X2	-0.11919	0.24486	-0.487	0.633

Residual standard error: 1.017 on 17 degrees of freedom
Multiple R-squared: 0.02204, Adjusted R-squared: -0.09301
F-statistic: 0.1916 on 2 and 17 DF, p-value: 0.8274

Going to higher dimensions →

```
1 n <- 20 # number of samples
2 p <- 10 # number of features / dimensions
3 Y <- rnorm(n)
4 X <- matrix(rnorm(n * p), n, p)
5 summary(lm(Y ~ X))
```

Call:
lm(formula = Y ~ X)

Residuals:

	Min	1Q	Median	3Q	Max
	-1.0255	-0.4320	0.1056	0.4493	1.0617

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.54916	0.26472	2.075	0.0679 .
X1	0.30013	0.21690	1.384	0.1998
X2	0.68053	0.27693	2.457	0.0363 *
X3	-0.10675	0.26010	-0.410	0.6911
X4	-0.21367	0.33690	-0.634	0.5417
X5	-0.19123	0.31881	-0.600	0.5634
X6	0.81074	0.25221	3.214	0.0106 *
X7	0.09634	0.24143	0.399	0.6992
X8	-0.29864	0.19004	-1.571	0.1505
X9	-0.78175	0.35408	-2.208	0.0546 .
X10	0.83736	0.36936	2.267	0.0496 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8692 on 9 degrees of freedom
Multiple R-squared: 0.6592, Adjusted R-squared: 0.2805
F-statistic: 1.741 on 10 and 9 DF, p-value: 0.2089

Going to even higher dimensions →

```
1 n <- 20 # number of samples
2 p <- 20 # number of features / dimensions
3 Y <- rnorm(n)
4 X <- matrix(rnorm(n * p), n, p)
5 summary(lm(Y ~ X))
```

Call:
lm(formula = Y ~ X)

Residuals:
ALL 20 residuals are 0: no residual degrees of freedom!

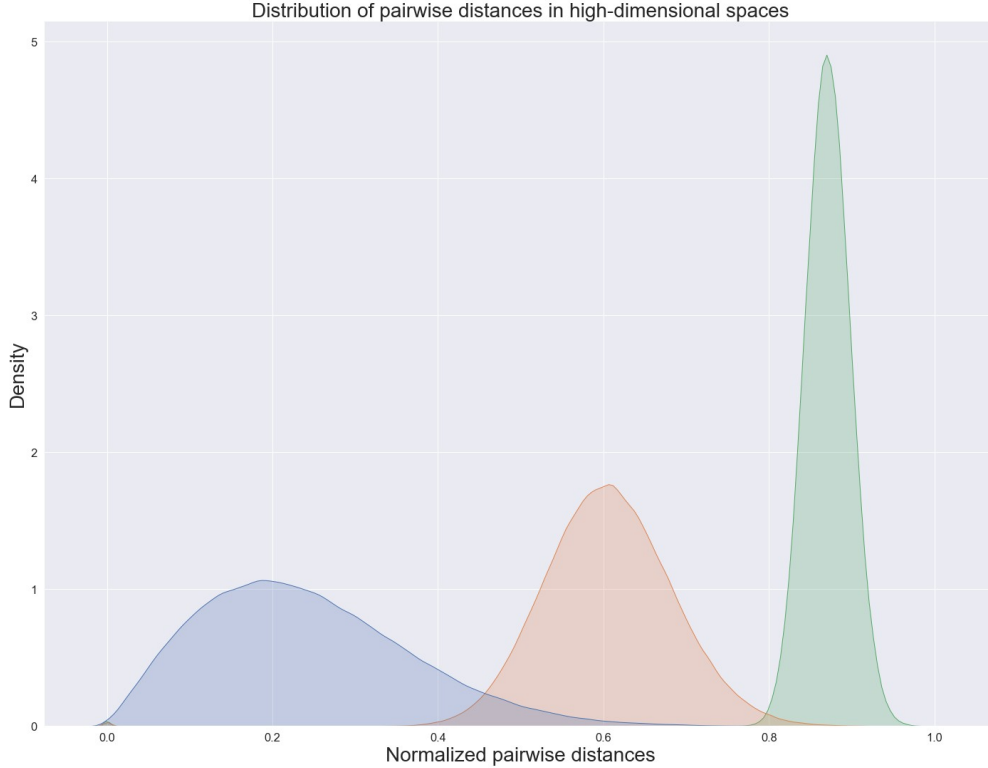
Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.34889	NaN	NaN	NaN
X1	0.66218	NaN	NaN	NaN
X2	0.76212	NaN	NaN	NaN
X3	-1.35033	NaN	NaN	NaN
X4	-0.57487	NaN	NaN	NaN
X5	0.02142	NaN	NaN	NaN
X6	0.40290	NaN	NaN	NaN
X7	0.03313	NaN	NaN	NaN
X8	-0.31983	NaN	NaN	NaN
X9	-0.92833	NaN	NaN	NaN
X10	0.18091	NaN	NaN	NaN
X11	-1.37618	NaN	NaN	NaN
X12	2.11438	NaN	NaN	NaN
X13	-1.75103	NaN	NaN	NaN
X14	-1.55073	NaN	NaN	NaN
X15	0.01112	NaN	NaN	NaN
X16	-0.50943	NaN	NaN	NaN
X17	-0.47576	NaN	NaN	NaN
X18	0.31793	NaN	NaN	NaN
X19	1.43615	NaN	NaN	NaN
X20	NA	NA	NA	NA

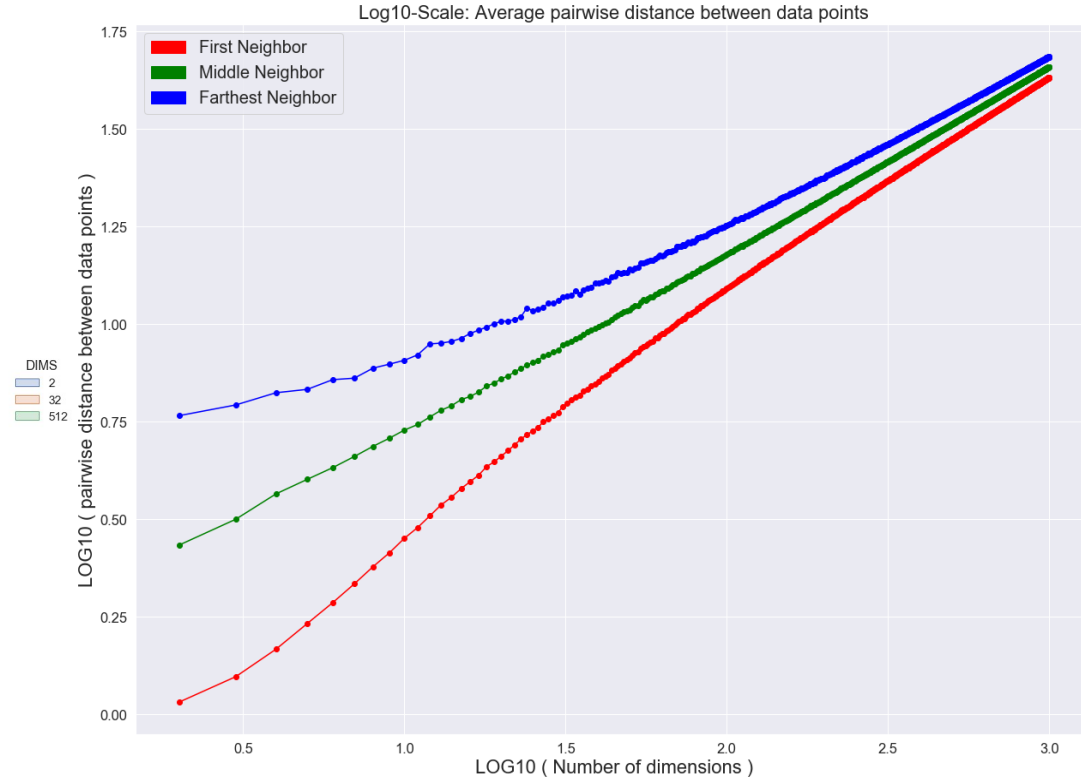
Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 19 and 0 DF, p-value: NA

This is another way we face the Curse of Dimensionality in computational biology

More on the Curse of Dimensionality

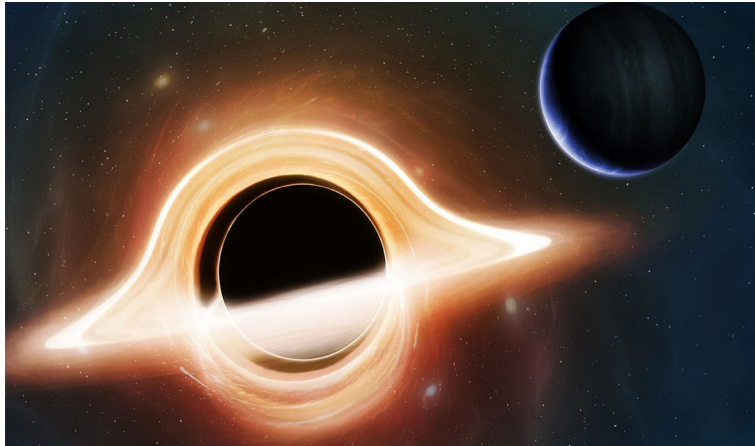
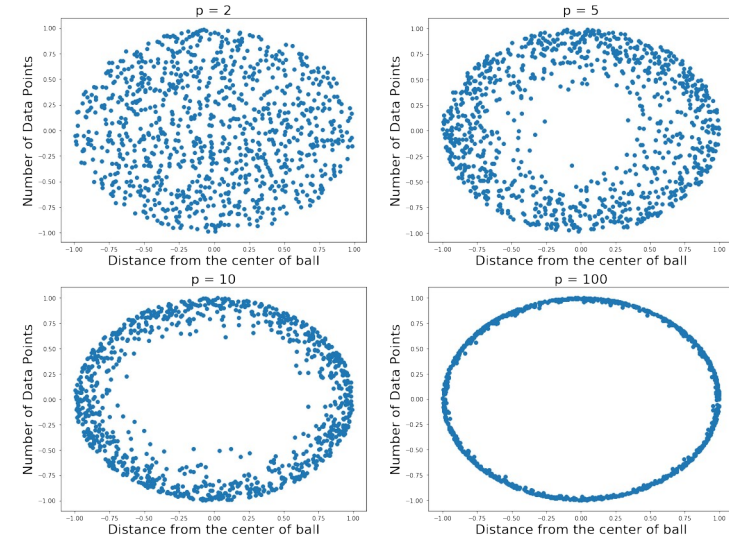
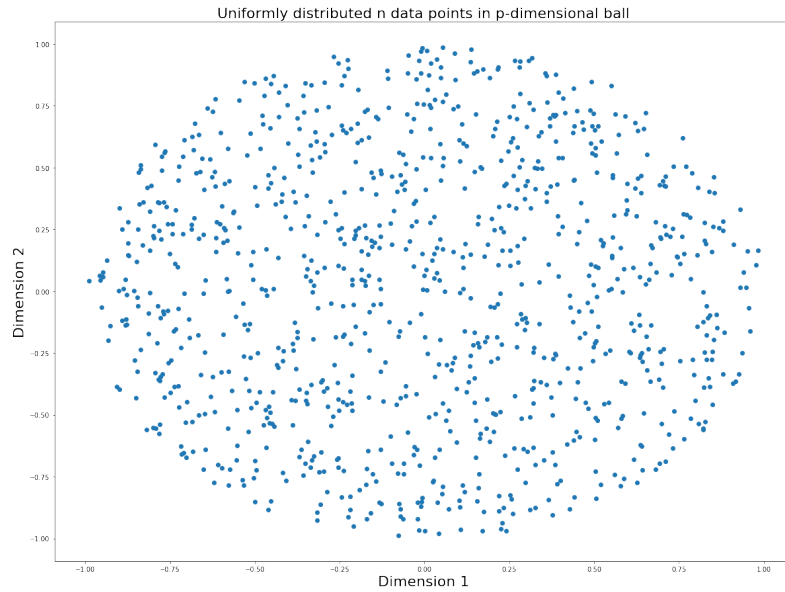


Data points become far from each other and equidistant in high dimensions



The differences between closest and farthest data point neighbours disappears in high-dimensional spaces: can't run cluster analysis

In high-dimensional space we can not separate cases and controls any more



High-dimensional data can be viewed as having a **“hole in the middle”**, hence the concept of mean / centroid loses its validity, hence we can't use Gaussian distribution

The curse(s) of dimensionality

There is such a thing as too much of a good thing.

Naomi Altman and Martin Krzywinski

We generally think that more information is better than less. However, in the 'big data' era, the sheer number of variables that can be collected from a single sample can be problematic. This embarrassment of riches is called the 'curse of dimensionality'¹ (CoD) and manifests itself in a variety of ways. This month, we discuss four important problems of dimensionality as it applies to data sparsity^{1,2}, multicollinearity³, multiple testing⁴ and overfitting⁵. These effects are amplified by poor data quality, which may increase with the number of variables.

Throughout, we use n to indicate the sample size from the population of interest and p to indicate the number of observed variables, some of which may have missing values for some samples. For example, we may have $n = 1,000$ subjects and $p = 200,000$ single-nucleotide polymorphisms (SNPs).

First, as the dimensionality p increases, the 'volume' that the samples may occupy grows rapidly. We can think of each of the n

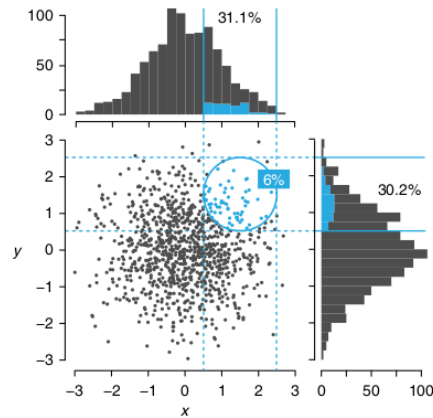


Fig. 1 | Data tend to be sparse in higher dimensions. Among 1,000 (x, y) points in which both x and y are normally distributed with a mean of 0 and s.d. $\sigma = 1$, only 6% fall within σ of $(x, y) = (1.5, 1.5)$ (blue circle). However, when the data are projected into a lower dimension—shown by histograms—about 30% of the points (all bins within blue solid lines) are within σ of 1.5. Blue

A and 100 to have the minor allele a. If we tabulate on two SNPs, A and B, we will expect only ten samples to exhibit both minor alleles with genotype ab. With SNPs A, B and C, we expect only one sample to have genotype abc, and with four or more SNPs, we expect empty cells in our table. We need a much larger sample size to observe samples with all the possible genotypes. As p increases, we may quickly find that there are no samples with similar values of a predictor.

Even with just five SNPs, our ability to predict and classify the samples is impeded because of the small number of subjects that have similar genotypes. In situations where there are many gene variants, this effect is exacerbated, and it may be very difficult to find affected subjects with similar genotypes and hence to predict or classify on the basis of genetic similarity.

If we treat the distance between points (e.g., Euclidian distance) as a measure of similarity, then we interpret greater distance as greater dissimilarity. As n increases, this

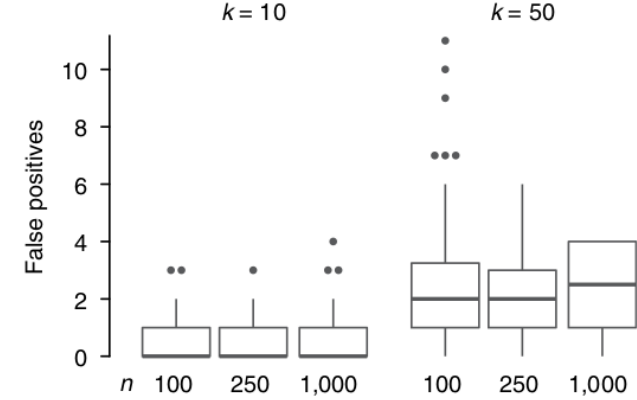


Fig. 3 | The number of false positives increases with each additional predictor. The box plots show the number of false positive regression-fit P values (tested at $\alpha = 0.05$) of 100 simulated multiple regression fits on various numbers of samples ($n = 100, 250$ and $1,000$) in the presence of one true predictor and $k = 10$ and 50 extraneous uncorrelated predictors. Box plots show means (black center lines), 25th and 75th percentiles (box edges), and minimum and maximum values (whiskers). Outliers (dots) are jittered.

Correcting for multiple testing does not solve the problem of too many false-positive hits

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metabolomics
N $\approx$ P

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Proteomics

$N \approx P$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metagenomics

- manageable

$$\mathbf{N} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$$

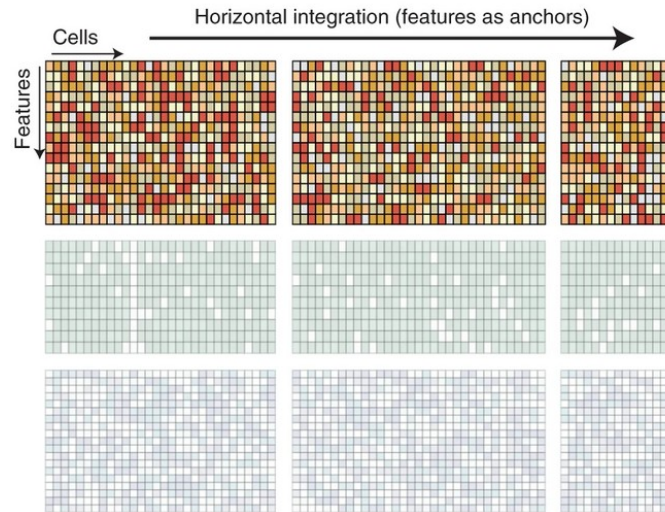
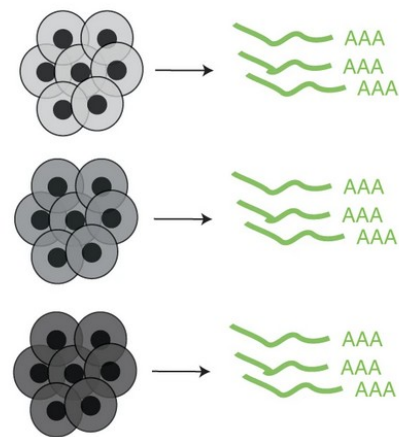
Transcriptomics
 $N \ll P$
 (Single cell: $N \leq P$)

challenging

$$\begin{aligned}
 & \text{Genomics} \\
 & N \llll P \\
 & \text{Methylomics} \\
 & N \llll P
 \end{aligned}$$

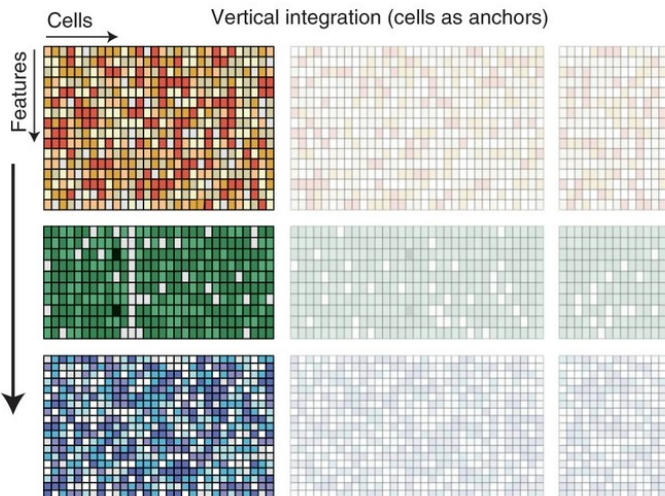
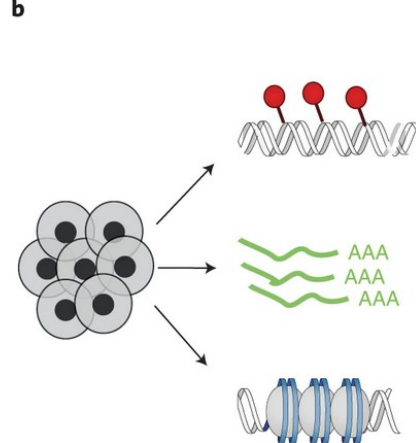
Multi-Omics Data Integration

a



Similar to batch correction problem

b



Here I will focus on integration across features

Statistical observations:
e.g. samples, cells etc.

Features: genes, proteins,
microbes, metabolites etc.

P_1

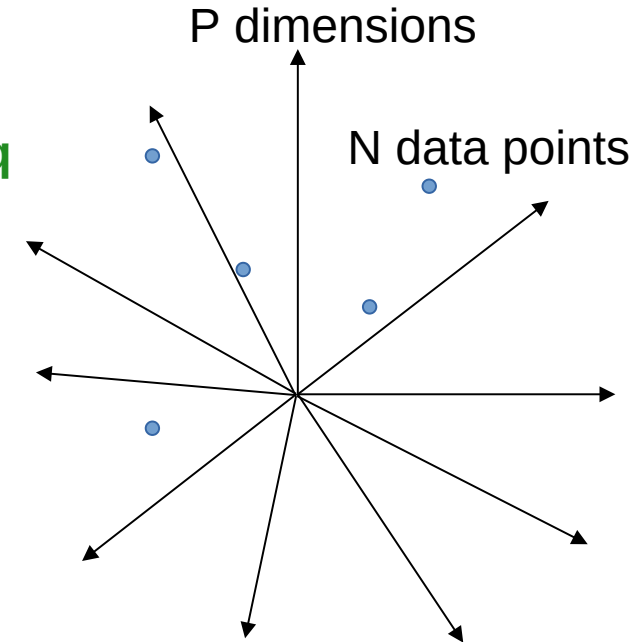
$$\begin{matrix}
 & & & & & & & & & \text{N} \\
 \begin{pmatrix}
 0 & 3 & 1 & 0 & 2 & 3 & 8 & 1 & 1 & 3 \\
 1 & 1 & 0 & 0 & 7 & 1 & 2 & 2 & 3 & 3 \\
 1 & 2 & 2 & 0 & 0 & 6 & 7 & 1 & 2 & 2 \\
 1 & 2 & 3 & 10 & 0 & 4 & 6 & 1 & 0 & 5 \\
 3 & 2 & 2 & 1 & 4 & 3 & 2 & 1 & 6 & 0 \\
 7 & 4 & 4 & 5 & 3 & 9 & 6 & 1 & 6 & 1 \\
 7 & 1 & 1 & 5 & 2 & 8 & 9 & 1 & 3 & 6 \\
 5 & 0 & 1 & 6 & 2 & 0 & 0 & 0 & 1 & 5 \\
 1 & 6 & 3 & 3 & 4 & 6 & 2 & 0 & 1 & 1 \\
 1 & 2 & 2 & 4 & 1 & 1 & 3 & 0 & 8 & 2
 \end{pmatrix}
 \end{matrix}$$

P_2

$$\begin{matrix}
 & & & & & & & & & \text{N} \\
 \begin{pmatrix}
 0 & 3 & 1 & 0 & 2 & 3 & 8 & 1 & 1 & 3 \\
 1 & 1 & 0 & 0 & 7 & 1 & 2 & 2 & 3 & 3 \\
 1 & 2 & 2 & 0 & 0 & 6 & 7 & 1 & 2 & 2 \\
 1 & 2 & 3 & 10 & 0 & 4 & 6 & 1 & 0 & 5 \\
 3 & 2 & 2 & 1 & 4 & 3 & 2 & 1 & 6 & 0 \\
 7 & 4 & 4 & 5 & 3 & 9 & 6 & 1 & 6 & 1 \\
 7 & 1 & 1 & 5 & 2 & 8 & 9 & 1 & 3 & 6 \\
 5 & 0 & 1 & 6 & 2 & 0 & 0 & 0 & 1 & 5 \\
 1 & 6 & 3 & 3 & 4 & 6 & 2 & 0 & 1 & 1 \\
 1 & 2 & 2 & 4 & 1 & 1 & 3 & 0 & 8 & 2
 \end{pmatrix}
 \end{matrix}$$

Omic1:
RNAseq

Omic2:
BSseq

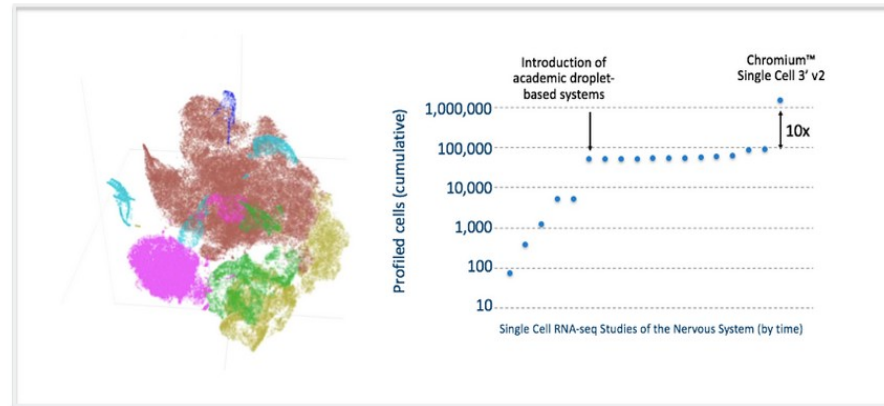


$P_1 + P_2 \gg N$ integration across features leads to even more high-dimensional data



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< Newer Article Older Article >



Our 1.3 million single cell dataset is ready to download



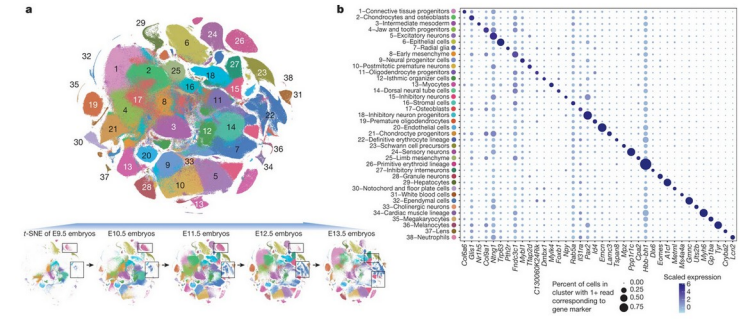
POSTED BY: **grace-10x**, on Feb 21, 2017 at 2:28 PM

At ASHG last year, we announced our 1.3 Million Brain Cell Dataset, which is, to date, the largest dataset published in the single cell RNA-sequencing (scRNA-seq) field. Using the Chromium™ Single Cell 3' Solution (v2 Chemistry), we were able to sequence and profile 1,308,421 individual cells from embryonic mice brains. Read more in our application note [Transcriptional Profiling of 1.3 Million Brain Cells with the Chromium™ Single Cell 3' Solution](#).



Fig. 2: Identifying the major cell types of mouse organogenesis.

From: *The single-cell transcriptional landscape of mammalian organogenesis*



a, t-SNE visualization of 2,026,641 mouse embryo cells (after removing a putative doublet cluster), coloured by cluster identity (ID) from Louvain clustering (in b), and annotated on the basis of marker genes. The same t-SNE is plotted below, showing only cells from each stage (cell numbers from left to right: $n = 151,000$ for E9.5; 370,279 for E10.5; 602,784 for E11.5; 468,088 for E12.5; 434,490 for E13.5). Primitive erythroid (transient) and definitive erythroid (expanding) clusters are boxed. b, Dot plot showing expression of one selected marker gene per cell type. The size of the dot encodes the percentage of cells within a cell type in

Solutions
Resources

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A next-generation platform to re-analyze published single-cell sequencing data

Single Cell Analysis

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September

by biomembers • August 30, 2019

Human Cell Atlas, single-cell data

We are glad to announce that we will upsize the current single-cell database in [BioTuring Single-cell Browser](#) to 5,500,000 cells this September. With this release, we will double the current number of publications indexed in BioTuring Single-cell Browser, and cross the number of cells hosted on available public single-cell data repositories like [Human Cell Atlas \(HCA\)](#) and [Broad Institute's Single-cell Portal](#).

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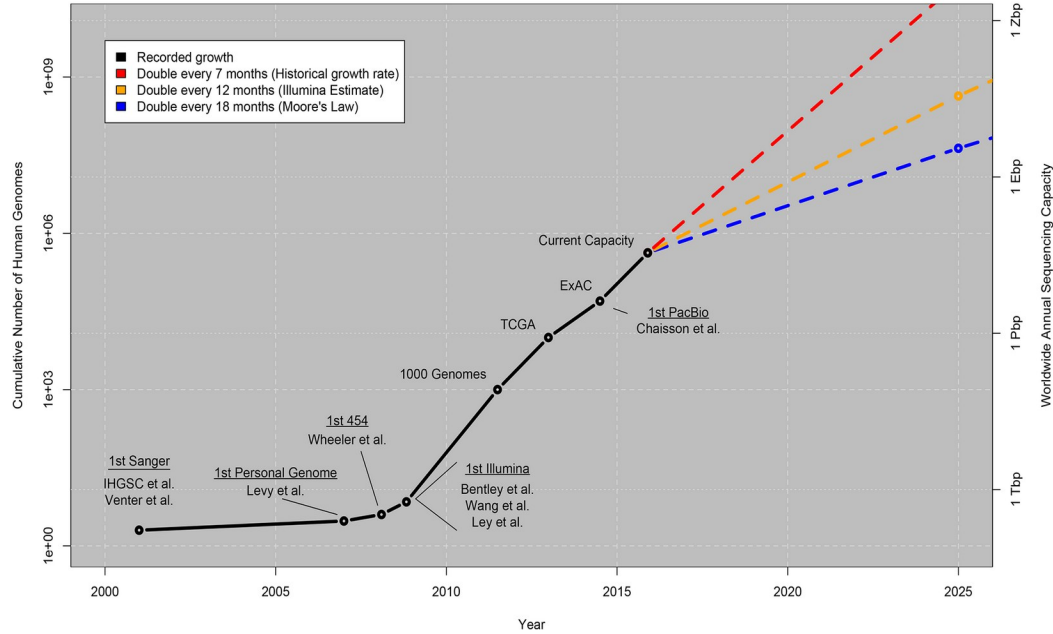
A new tool to interactively visualize single-cell objects (Seurat, Scanpy, SingleCellExperiments, ...)

September 26, 2019

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September

August 30, 2019

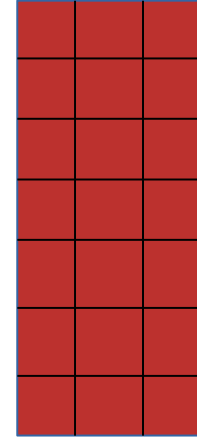
Growth of DNA Sequencing



Stephens et al., (2015). Big Data: Astronomical or Genomical? PLoS Biology 13(7)

Genomics / WGS: Little Data

$$N_1 \sim 10^3$$

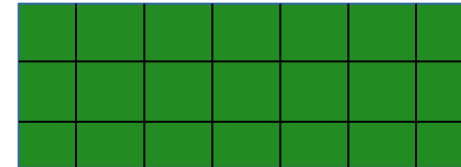


$$P_1 \sim 10^6$$

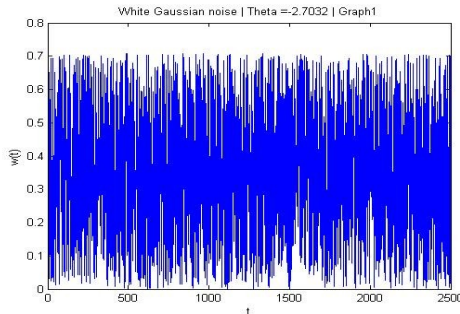
$$N_1 * P_1 = N_2 * P_2 = 10^9$$

scRNAseq: Big Data

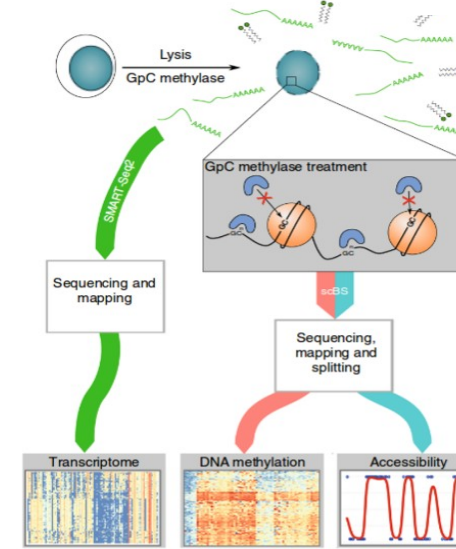
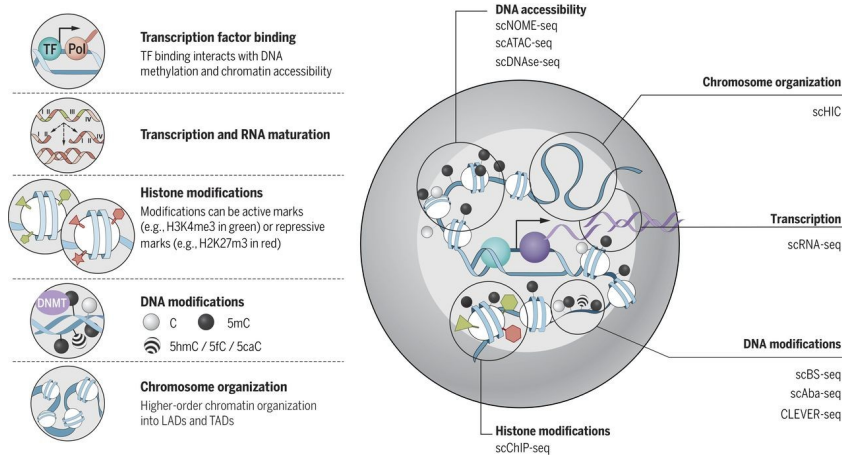
$$N_2 \sim 10^6$$



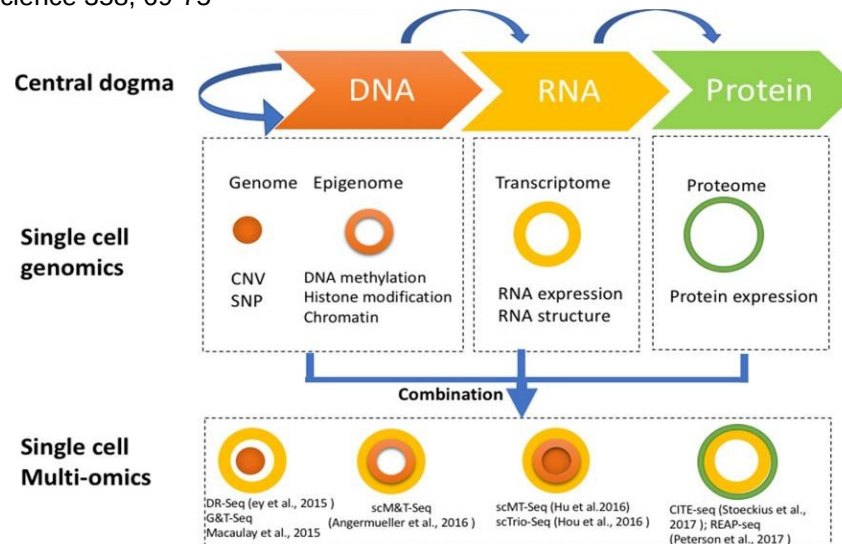
$$P_2 \sim 10^3$$



A file with **White Noise** can also take a lot of disk space



Clark et al., 2018, Nature Communications 9, 781



Kelsey et al., 2017, Science 358, 69-75

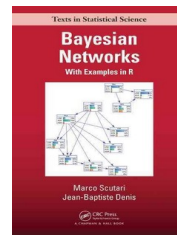
Hu et al., 2018, Frontier in Cell and Developmental Biology 6, 1-13

How to define and evaluate
multi-Omics data integration?

OnPLS

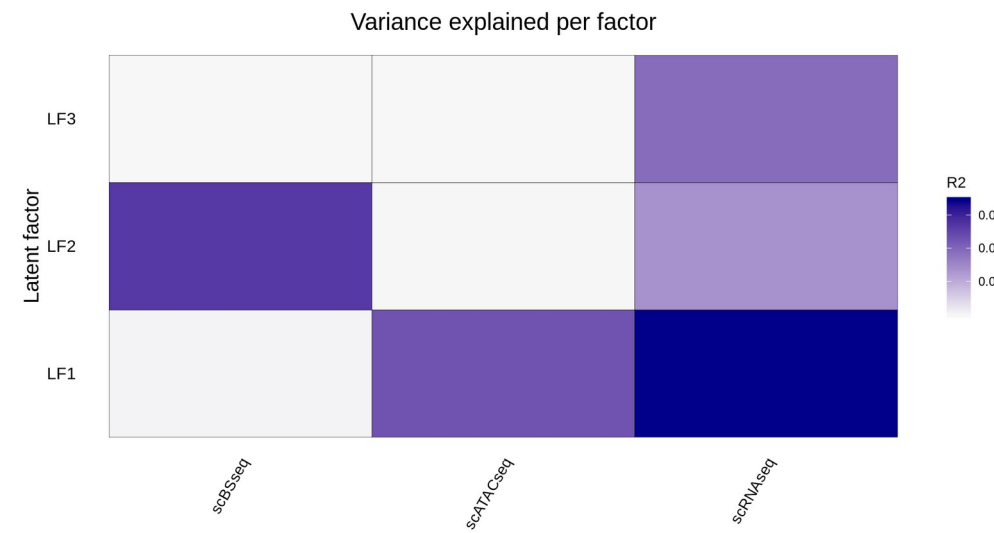
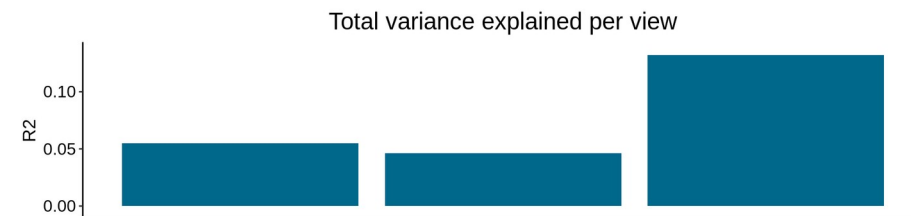
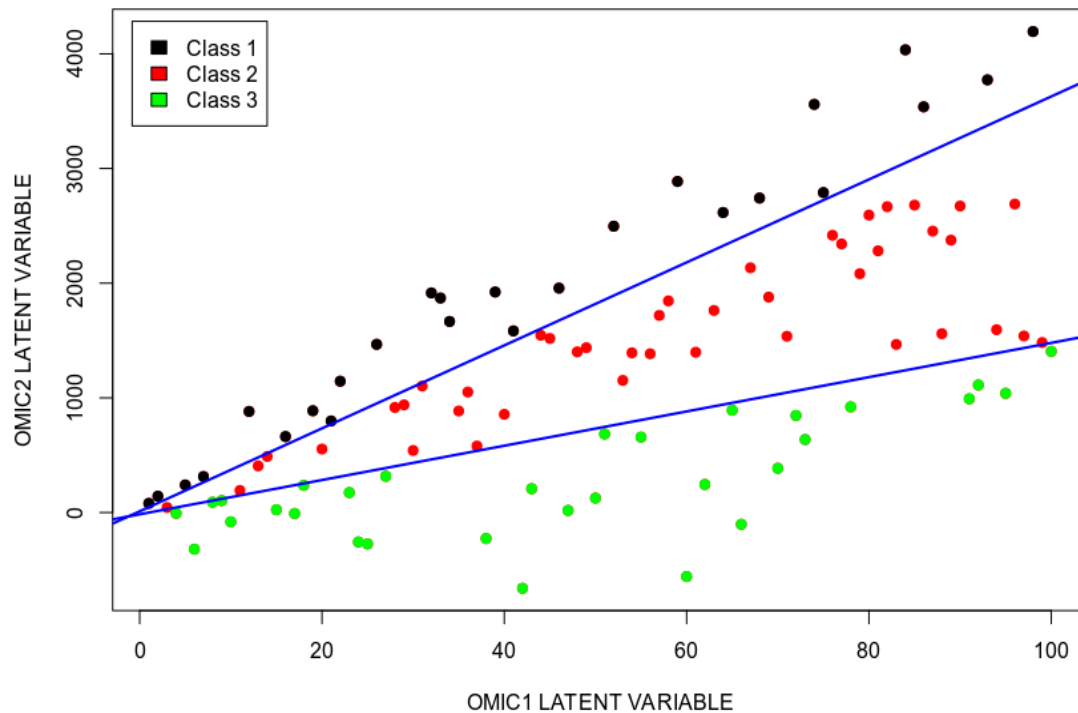
JIVE

DISCO



Clustering of Clusters

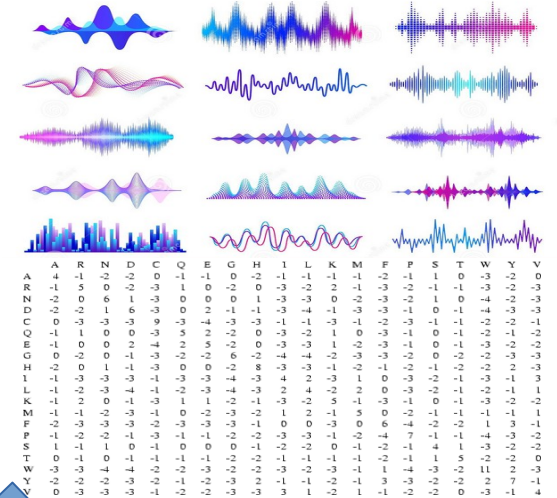
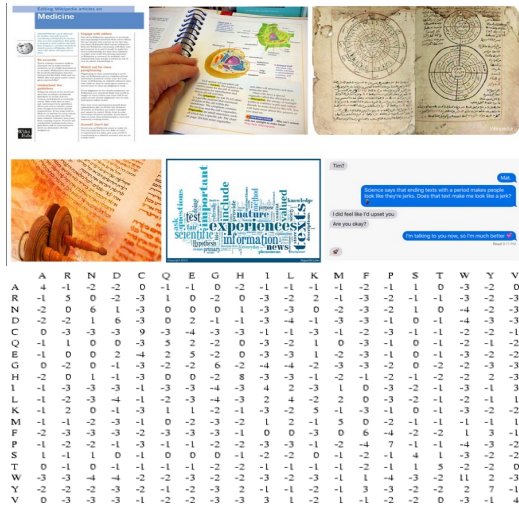
**Idea Behind Omics Integration:
See Patterns Hidden in Individual Omics**



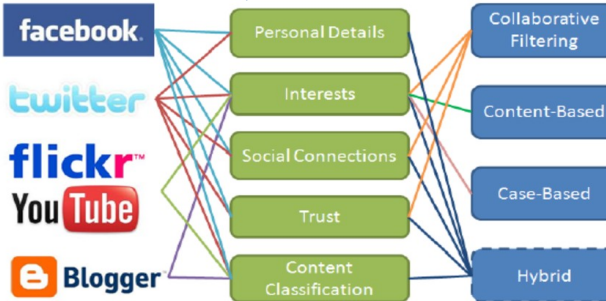
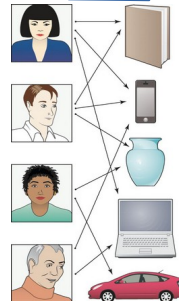
TEXT (78%)

IMAGE (83%)

SOUND (75%)



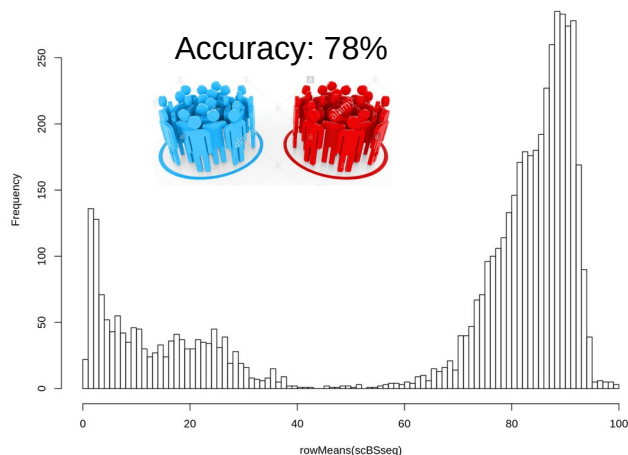
Predict Facebook user interests



Data Integration Accuracy: 96%

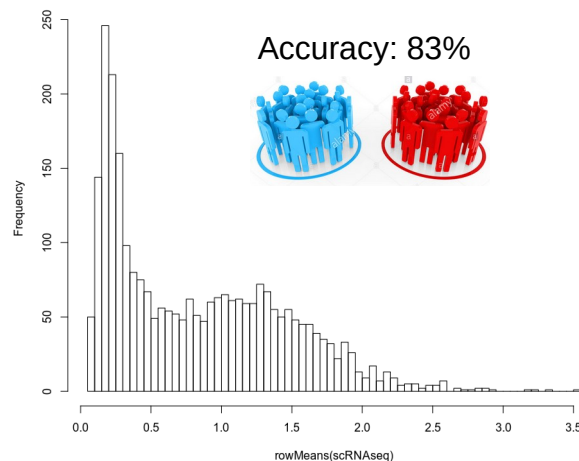
Methylation

scBSseq



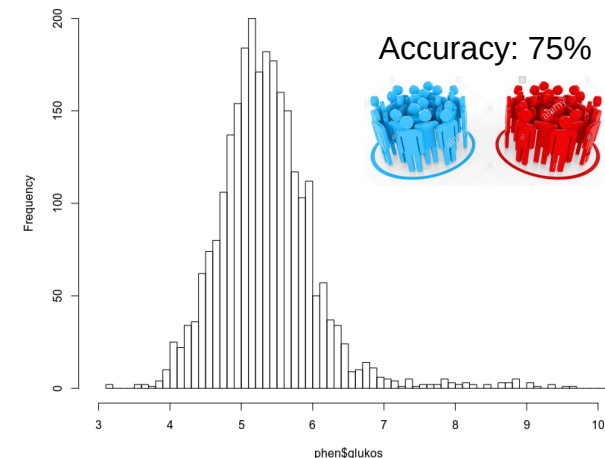
Gene Expression

scRNAseq



Clinical variable

Phenotype



1) Convert to common space:

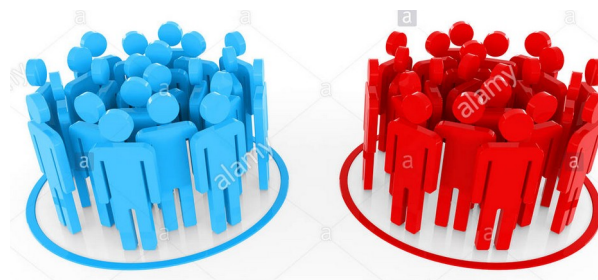
Neural Networks, SNF, UMAP

2) Explicitly model distributions:

MOFA, Bayesian Networks

3) Extract common variation:

PLS, CCA, Factor Analysis



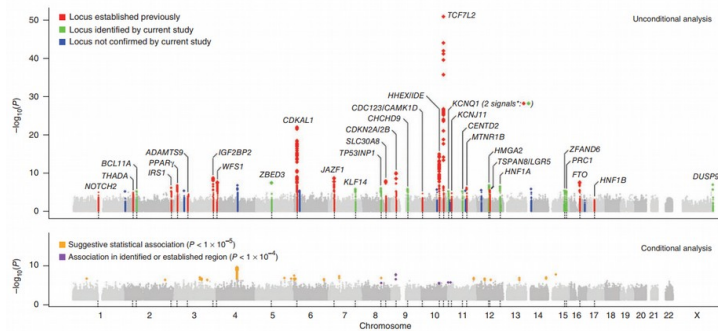
HEALTHY

SICK

Data Integration

Accuracy: 96%

Statistics searches for candidates



Consequence



NEWS FEATURE PERSONAL GENOMES NATURE Vol 456/6 November 2008



The case of the missing heritability

B. Maher, Nature 456, 18-21 (2008)

Machine Learning optimizes prediction



Letter | Published: 31 July 2019

A clinically applicable approach to continuous prediction of future acute kidney injury

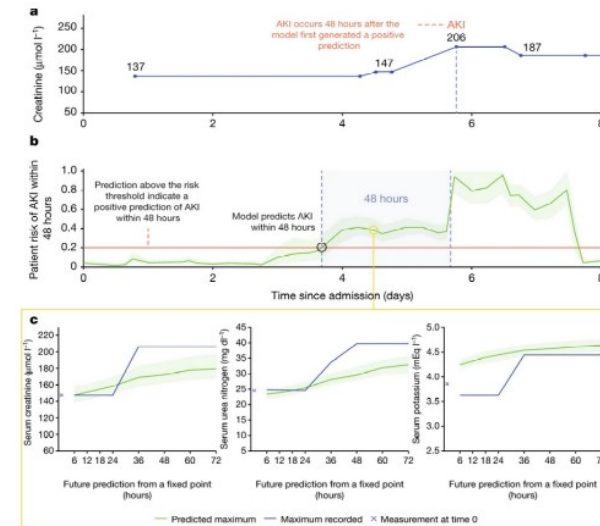
Renad Tomalov, Xavier Glorot, [...] Shaker Mohamed

Nature 572, 116-119 (2019) | Download Citation

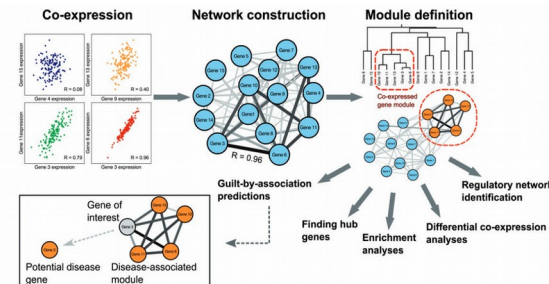
Abstract

The early prediction of deterioration could have an important role in supporting healthcare professionals, as an estimated 11% of deaths in hospital follow a failure to promptly recognize and treat deteriorating patients¹. To achieve this goal requires predictions of patient risk that are continuously updated and accurate, and delivered at an individual level with sufficient context and enough time to act. Here we develop a deep learning approach for the continuous risk prediction of future deterioration in patients, building on recent work that models adverse events from electronic health records^{2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17} and using acute kidney injury—a common and potentially life-threatening condition¹⁸—as an exemplar. Our model was developed on a large, longitudinal dataset of electronic health records that cover diverse

From: A clinically applicable approach to continuous prediction of future acute kidney injury



Consequence



- 1) Biological data are **high-dimensional** and notoriously difficult to analyze
- 2) Integration across Omics is often sensitive to the **Curse of Dimensionality**
- 3) Integrating across Omics we expect to discover **novel patterns** in the data
- 4) Increased **prediction accuracy** is an indication of successful data integration
- 5) **Single cell Omics** are promising for integration in terms of statistical power



*Knut och Alice
Wallenbergs
Stiftelse*



LUNDS
UNIVERSITET